

《计算机网络》作业

第十次作业：（英文版教材第五章 23，24，25，27，28，29，30，31，38，补充题 2）

23. (5.6.2 IP Addresses) Suppose that instead of using 16 bits for the network part of a class B address originally, 20 bits had been used. How many class B networks would there have been?
24. (5.6.2 IP Addresses) Convert the IP address whose hexadecimal representation is C22F1582 to dotted decimal notation.
25. (5.6.2 IP Addresses) A network on the Internet has a subnet mask of 255.255.240.0. What is the maximum number of hosts it can handle?
27. (5.6.2 IP Addresses-Classless Addressing) A large number of consecutive IP addresses are available starting at 198.16.0.0. Suppose that four organizations, A, B, C, and D, request 4000, 2000, 4000, and 8000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation.
28. (5.6.2 IP Addresses-Classless Addressing) A router has just received the following new IP addresses: 57.6.96.0/21, 57.6.104.0/21, 57.6.112.0/21, and 57.6.120.0/21. If all of them use the same outgoing line, can they be aggregated? If so, to what? If not, why not?
29. (5.6.2 IP Addresses-Classless Addressing) The set of IP addresses from 29.18.0.0 to 29.18.127.255 has been aggregated to 29.18.0.0/17. However, there is a gap of 1024 unassigned addresses from 29.18.60.0 to 29.18.63.255 that are now suddenly assigned to a host using a different outgoing line. Is it now necessary to split up the aggregate address into its constituent blocks, add the new block to the table, and then see if any reaggregation is possible? If not, what can be done instead?
30. (5.6.2 IP Addresses-Routing Table) A router has the following (CIDR) entries in its routing table:

Network address	Next hop/Interface
135.46.56.0/22	Interface 0
135.46.60.0/22	Interface 1
192.53.40.0/23	Router 1
0.0.0.0/0 (default)	Router 2

For each of the following IP addresses, what does the router do if a packet with that address arrives?

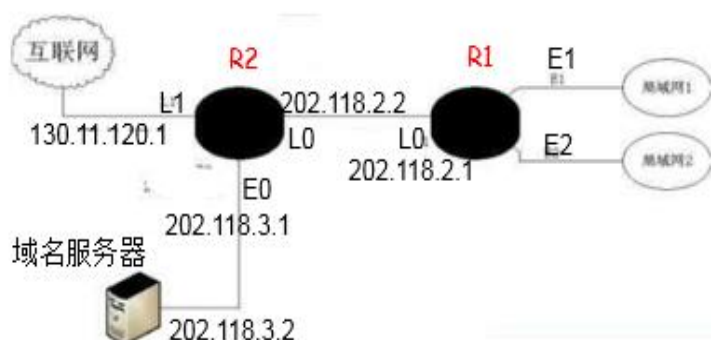
- (a) 135.46.63.10
- (b) 135.46.57.14
- (c) 135.46.52.2
- (d) 192.53.40.7
- (e) 192.53.56.7

31. (5.6.2 IP Addresses-NAT) Many companies have a policy of having two (or more) routers connecting the company to the Internet to provide some redundancy in case one of them goes down. Is this policy still possible with NAT? Explain your answer.

38. (5.6.3 IPv6) The *Protocol* field used in the IPv4 header is not present in the fixed IPv6 header. Why not?

补充题 2(5.6.2 IP Addresses):

某公司网络拓扑图如下图所示，路由器 R1 通过接口 E1，E2 分别连接局域网 1，局域网 2，通过接口 L0 连接路由器 R2，并通过路由器 R2 连接域名服务器与互联网。



R1 和 R2 的路由表结构为：

目的网络 IP 地址	子网掩码	下一跳 IP 地址	接口
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1) 将 IP 地址空间 202.118.1.0/24 划分为两个子网，分配给局域网 1 和局域网 2，其中局域网 1 可分配的地址数不少于 120 个，局域网 2 可分配的地址数不少于 30 个。请按照最小够用原则给出两个子网划分

结果（见表 1）。说明理由或给出必要的计算过程。

	网络地址	子网掩码	可用的最小地址	可用的最大地址
局域网 1				
局域网 2				

2) 请给出 R1 的路由表，使其明确包括到局域网 1 的路由，局域网 2 的路由，域名服务器的主机路由和互联网的路由。请采用路由聚合技术，给出 R2 到局域网 1 和局域网 2 的路由。